



NATIONAL UNIVERSITY
of Computer & Emerging Sciences

Gaze Detection for Online Assessments /Driving

COURSE:

Deep Learning for Perception

INSTRUCTOR:

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MEMBERS:

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1. Project Objective:

The primary objective of this project was to build a gaze detection system for use in online assessments and driving monitoring scenarios. In the context of online assessments, the tool aims to enhance academic integrity by detecting when a student diverts their gaze away from the screen—potentially indicating distraction or dishonest behavior. In driving contexts, gaze monitoring can help identify driver inattentiveness or drowsiness.

The system leverages computer vision and deep learning techniques to process webcam footage in real time and classify the direction of a user's gaze into three categories: Left, Right, and Center.

2. Datasets Used:

• Custom Real-World Images Dataset

To tailor the system for realistic environments such as online assessments, a custom dataset was created. Group members captured labeled images representing different gaze directions (left, right, center) under controlled conditions. Given the limited number of images, extensive data augmentation techniques were applied, including:

- a) Rotation

- b) Zoom
- c) Contrast adjustment
- d) Brightness variation

These augmentations helped create a more generalized and sparse dataset suitable for training deep learning models.

• **Challenges in Dataset Collection**

Most publicly available datasets consisted of either synthetically generated 3D images or unbalanced samples, which negatively impacted training.

Datasets with only single-eye images were not compatible with the designed model, which required both eyes and facial landmarks for accurate prediction.

As a result, training with external data led to poor generalization and incorrect gaze classification.

3. Models Used:

The system was developed using a combination of traditional image processing and modern deep learning techniques.

• **Challenges in Model Development**

Due to the lack of high-quality data early on, initial models built from scratch underperformed. These early attempts struggled with accuracy and consistency due to poor feature learning.

• **EfficientNetB0**

The final and most effective model used was EfficientNetB0, a lightweight but high-performing CNN architecture developed by Google. It belongs to the EfficientNet family, known for its compound scaling strategy that balances depth, width, and resolution efficiently.

The pre-trained EfficientNetB0 model was fine-tuned on our custom dataset using:

ReduceLROnPlateau

EarlyStopping

ModelCheckpoint

These callbacks helped prevent overfitting and ensured better generalization.

• **Eye Region Extraction**

Facial landmarks were detected using classical computer vision techniques. Bounding boxes were placed around the eyes using landmark coordinates to isolate eye regions for gaze prediction.

- **Custom CNN for Gaze Classification**

An alternative CNN architecture was also tested, with the following structure:

Convolutional layers for extracting spatial features

MaxPooling layers to reduce dimensionality

Fully connected layers for final classification

Softmax output layer with three neurons corresponding to Left, Right, and Center

4. Results:

The final model yielded strong performance on the validation dataset:

Accuracy: 91%

Precision / Recall:

Left: 90% / 89%

Right: 92% / 91%

Center: 91% / 92%

Inference Speed: Real-time (approximately 25 FPS on CPU)

Confusion matrices and example predictions showed the model's ability to generalize well and highlighted occasional misclassifications under poor lighting or unusual head angles.

It demonstrated great results on test and validation data , but when tested on some new unseen images it gave a decent accuracy of upto 70% which can be further improved by more fine tuning and generalizing the dataset.

5. Discussion on Results

The system demonstrated promising results and appears suitable for real-time deployment in both online proctoring . However, certain challenges and limitations remain:

- **Lighting Conditions**

Prediction accuracy was sensitive to variations in lighting.

Image normalization and augmentation helped mitigate this, but further robustness is needed.

- **Head Movement**

The model performs reliably with minor head movement but tends to misclassify when users tilt their head significantly or look down.

- **User Diversity**

Differences in eye shape, skin tone, or glasses affected performance.

A more diverse training dataset would improve fairness and generalization.

- **Dataset Quality**

The current dataset, while useful, is limited in size and variety.

A larger, real-world dataset from exam or driving scenarios would significantly enhance the system's performance and reliability.